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Velocity-Based Resistance Training in Soccer: Practical Applications and Technical Considerations

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ABSTRACT

Since 2010, the velocity-based approach (VBT) has been established as a reliable method to be used during resistance training (RT) interventions. VBT lies in the monitoring of the barbell velocity during the concentric phase of an exercise, which is a direct consequence of the force applied by the athlete to the external load. VBT offers coaches and physical trainers of soccer teams a wide spectrum of practical applications. Considering the close association between the barbell velocity and the relative intensity (i.e., load-velocity relationship), practitioners could ensure “on the go” that players train very close to the programmed load in each training set. Methodologies derived from the VBT, such as the velocity loss or the “level of effort,” allow to plan and control the intraset fatigue to be incurred by the

soccer players during RT. Incorporating the barbell velocity monitoring into the testing routine would permit the determination of the changes in strength across the entire spectrum of loads and offer a valid and practical proposal to evaluate the neuromuscular recovery status. The current article aimed to review in-depth the above-mentioned practical applications and some technological and technical aspects to consider when implementing the VBT in soccer.

INTRODUCTION

Most players' actions during a soccer game involve very rapid contractions, thus the time available to produce force is very limited. Hence, the strength capacity and particularly the ability to produce force quickly is a critical part of optimizing physical performance (58). Numerous studies have demonstrated that resistance training (RT) programs transfer positively to common actions

in soccer matches, such as linear sprints (55,57), changes of direction (17,18), and vertical jumps (12,44). However, managing RT sessions within a soccer team schedule is a challenge considering that players are dealing with weekly competitions. Accumulated fatigue may impair gains in strength and likely increase the injury risk if RT is not properly prescribed (6,64). To reduce hazards, it seems more than advisable to track players' performance status and use this information to adapt the training process according to the players' aims, responses, and evolution.

Advances in sport sciences have led to the development of technology-based methods and procedures to assist coaches in managing RT parameters. Particularly, velocity-based training (VBT) is attracting attention in the

KEY WORDS:

football; conditioning; barbell velocity; testing; team sports; speed

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team sports community, given its potential advantages to adjust training intensity and maximize athletic performance (62). In this article, we viewed in-depth the practical applications of VBT, its limitations and the technological and technical aspects to consider when implementing the VBT in soccer. In addition, we provided an excel sheet (see Supplementary material, Supplemental Digital Content, <http://links.lww.com/SCJ/A340>) to easily apply the VBT in the field for (a) determining the target velocity to be attained at the first repetition of the set (corresponding to the planned relative intensity, % 1RM), (b) determining the %1RM that is being used in the set based on the velocity attained, (c) elaborating an individual load-velocity (L-V) relationship for each player, and (d) evaluating the changes in dynamic strength across the entire spectrum of loads.

VELOCITY-BASED TRAINING AND RELATIVE INTENSITY (%1RM)

TRADITIONAL METHODOLOGIES

The configuration of training stimuli involves the manipulation of specific variables to achieve the targeted goals, including the %1RM (46). To date, most RT interventions involving soccer players have followed 2 traditional strategies to program the %1RM to be used throughout the training plan: (a) the absolute load (in kilograms) corresponding to a given percentage of the 1RM (kg-%1RM) and (b) the fixed number of repetitions to failure (*n*RM) (42,53,57). The first one requires an initial evaluation of the 1RM absolute load, from which the different %1RMs are programmed as a percentage of this absolute load. For example, a player with an initial 1RM of 100 kg in the full squat would use an absolute load of 60, 70, and 80 kg to train this exercise at his or her 60, 70, and 80% 1RM, respectively (3,8). Nevertheless, the 1RM value of a player varies daily as a result of changes in dynamic strength or motivation state, hence the kg-%1RM method lacks the accuracy to ensure that the target %1RM is equal to the actual % 1RM (39,46). The *n*RM method, for its

part, involves the player selecting an absolute load at which he or she can complete a certain number of repetitions to volitional fatigue (*n*), which would be theoretically associated with a specific % 1RM (e.g., 12RM = 75% 1RM) (9,60). However, the traditional *n*RM methodology itself implies performing each training set to muscle failure, which has proven to be a dangerous, inefficient, and even detrimental practice for soccer performance (39). Considering the basic principle of the *n*RM approach, other methodologies such as the “level of effort,” similar to the “repetitions in reserve” strategy, have emerged (20). This methodology has shown to be a precise, reliable, and practical approach for programming the %1RM and the intraset fatigue (20).

CONTRIBUTIONS OF THE VELOCITY-BASED TRAINING

One practical application provided by the VBT is the determination of players' individual L-V relationship, based on the close association between the barbell velocity and the %1RM (21,29,33,49). The L-V relationship ensures that the player trains at the programmed % 1RM in each training set, thus avoiding the meaningful mismatches that might occur when programming is based on the kg-%1RM method and the negative effects related to the traditional *n*RM approach (12,35,39,44). Briefly, the L-V relationship involves practitioners establishing a target mean propulsive velocity (MPV, mean velocity from the propulsive phase, defined as that portion of the concentric phase during which barbell acceleration [*a*] is greater than the acceleration because of gravity [$a \geq -9.81 \text{ m} \cdot \text{s}^{-2}$]) (48) to be attained in the fastest repetition (usually the first) of each training session. Thus, before starting each training set, the player should adjust the absolute load (in kilograms) placed on the bar to ensure that the MPV achieved at the beginning of the set matched that programmed ($\pm 0.03 \text{ m} \cdot \text{s}^{-1}$) (37,46).

Based on this assumption, practitioners could program the target % 1RM to be used in each resistance

exercise using its specific L-V relationship (Table 1). Similarly, the L-V relationship allows coaches and physical trainers to determine the %1RM being used by the player “on the go,” as soon as the first repetitions against any given absolute load are performed with maximal voluntary effort (Table 2). It is expected that athletes perform the fastest execution in the first repetition, and then progressively velocity slows down as a result of fatigue. However, coaches should be aware that the first repetition likely can be not the fastest because of lack of concentration, improper technique, and insufficient familiarization, thus making VBT adjustments inaccurate. Importantly, athletes must be familiarized with performing each repetition as fast as they can. This fact will not only allow the VBT to be properly implemented but also maximize the neuromuscular and functional adaptations produced after a RT intervention (11,36). Therefore, after familiarization with the correct execution of the exercise, soccer players would benefit from performing the concentric phase as fast as they can. On the other hand, factors like the range of motion (ROM) used (higher MPV as the ROM increases) (29) or the strategy used to transition from the concentric to the eccentric phase (i.e., with or without pause, higher MPV during nonpaused executions) (34) significantly influence the resulted MPV. Hence, if the evaluators do not control these factors, they could not be confident that changes in barbell velocity are because of actual strength changes and not to variations in the technical execution.

However, even though the general L-V relationship presented above has significantly increased the accuracy of the %1RM prescription (Table 1) and monitoring (Table 2), practitioners could go a step further by modeling the individual L-V relationship for each player. This procedure represents the most precise methodology to prescribe the %1RM because it allows us to avoid the slight but meaningful between-subject differences in the velocity attained to each %1RM (62). For

Table 1
Mean propulsive velocity (MPV, in $\text{m} \cdot \text{s}^{-1}$) attained with each relative intensity (%1RM) for different multi-joint RT exercises

Load (%1RM)	Bench press	Prone bench pull	Full squat (paused) ^a	Full squat (nonpaused)	Deadlift	Shoulder press	Pull-ups
30%	1.27–1.30	1.47–1.52	—	—	—	—	—
35%	1.18–1.22	1.40–1.45	—	—	—	—	—
40%	1.09–1.13	1.33–1.37	0.92–0.95	1.26–1.30	1.01–1.11	1.20–1.25	—
45%	1.00–1.04	1.25–1.30	0.87–0.91	1.20–1.23	0.96–1.04	1.11–1.15	—
50%	0.92–0.95	1.18–1.23	0.82–0.86	1.13–1.16	0.89–0.97	1.02–1.06	—
55%	0.84–0.87	1.11–1.15	0.77–0.81	1.05–1.09	0.82–0.90	0.94–0.98	0.76–0.92
60%	0.76–0.79	1.04–1.08	0.72–0.75	0.98–1.01	0.75–0.83	0.85–0.89	0.68–0.82
65%	0.68–0.71	0.97–1.01	0.67–0.70	0.91–0.93	0.68–0.76	0.76–0.80	0.60–0.72
70%	0.60–0.63	0.90–0.94	0.62–0.65	0.83–0.85	0.63–0.69	0.67–0.71	0.51–0.63
75%	0.52–0.55	0.83–0.87	0.57–0.59	0.75–0.77	0.56–0.62	0.59–0.62	0.43–0.53
80%	0.45–0.47	0.77–0.80	0.51–0.53	0.67–0.69	0.49–0.55	0.49–0.52	0.34–0.44
85%	0.37–0.39	0.70–0.74	0.46–0.48	0.59–0.60	0.43–0.47	0.41–0.44	0.25–0.35
90%	0.30–0.32	0.64–0.67	0.40–0.42	0.50–0.51	0.37–0.41	0.34–0.36	—
95%	0.23–0.25	0.57–0.60	0.35–0.38	0.41–0.42	0.31–0.33	0.26–0.28	—
100%	0.16–0.18	0.51–0.54	0.28–0.31	0.32–0.33	0.23–0.27	0.18–0.20	—
Adjust (R^2)	0.97	0.94	0.96	0.96	0.96	0.97	0.96

Data are 95% CI.

^aImposing a momentary pause for 2 seconds between the eccentric and concentric phases (31,34). Bench press and prone bench pull (49); full squat (paused) (29); full squat (nonpaused) (50); deadlift (33); shoulder press (21); pull-ups (51).

RT = resistance training.

Table 2
Equations to estimate the relative intensity (%1RM) from the mean propulsive velocity (MPV) for different multi-joint resistance training exercises

Exercise	Equations	Adjust (R^2)	SEE (%)
Bench press	$\%1RM = 11.2988 \times (MPV^2) - (78.05 \times MPV) + 113.04$	0.97	4.39
Prone bench pull	$\%1RM = 13.2596 \times (MPV^2) - (93.867 \times MPV) + 144.38$	0.95	5.90
Full squat (paused) ^a	$\%1RM = 4.468 \times (MPV^2) - (96.223 \times MPV) + 127.51$	0.96	—
Full squat (nonpaused)	$\%1RM = -5.961 \times (MPV^2) - (50.71 \times MPV) + 117$	0.95	4.02
Deadlift	$\%1RM = 19.472 \times (MPV^2) - (99.908 \times MPV) + 124.452$	0.94	6.29
Shoulder press	$\%1RM = 10.428 \times (MPV^2) - (73.583 \times MPV) + 114.46$	0.97	4.13
Pull-ups	$\%1RM = -53.472 MPV + 110.68$	0.96	3.15

^aImposing a momentary pause for 2 seconds between the eccentric and concentric phases to increase reliability (31,34). SEE = standard error of estimation. Bench press and prone bench pull (49); full squat (paused) (29); full squat (nonpaused) (50); deadlift (33); shoulder press (21); pull-ups (51).

example, whereas the MPV attained at the 60% 1RM in the free weight full squat is $\sim 0.92 \text{ m} \cdot \text{s}^{-1}$ (resulting from a general L-V relationship including subjects with different characteristics, Figure), individual MPVs at that specific %1RM could range from $0.79 \text{ m} \cdot \text{s}^{-1}$ (lowest) to $1.03 \text{ m} \cdot \text{s}^{-1}$ (highest). Thus, because $\sim 0.07 \text{ m} \cdot \text{s}^{-1}$ would represent $\sim 5\%$ 1RM in this exercise, the players with the lowest ($0.79 \text{ m} \cdot \text{s}^{-1}$) and highest ($1.03 \text{ m} \cdot \text{s}^{-1}$) MPV could be training against 69 and 52% 1RM, respectively, if a general MPV ($0.92 \text{ m} \cdot \text{s}^{-1}$) was used (52). Therefore, we strongly recommend individualizing the use of the L-V relationship to prescribe RT, and thus optimize its benefits.

VELOCITY-BASED TRAINING AND INTRASET FATIGUE

TRADITIONAL METHODOLOGIES

The level of fatigue incurred within each training set (i.e., intraset fatigue) has been proposed as one of the main determinants of the adaptations generated after an RT intervention. The intraset fatigue would be modulated by the number of repetitions performed within the set concerning the maximum number of repetitions that could be completed (i.e., distance to muscle failure) (47). In general, the nearer the set to muscular failure, the higher levels

of mechanical and metabolic fatigue (13,32,37). These acute levels of intraset fatigue would generate different physiological signals that would modulate the training effects (26).

For example, although the α -motor neuron innervating each muscle fiber seems to be the main cause of its initial typology (i.e., I, I-IIA, IIA, IIA-IIX, or IIX) (5), this fiber phenotype has been proved to be modified as a function of the fatigue produced in each training set (1,38). Likewise, the continuous exposition to high levels of blood ammonia and its associated pathways (purine cycle), generated when high intraset fatigue is incurred (47), could decrease the intramuscular adenine nucleotide content (19,56) and therefore the capability to produce force. Results found by 2 systematic review and meta-analysis studies revealed that submaximal levels of intraset fatigue would be more effective and efficient (lower total volume throughout the RT program) than maximal ones to produce optimal neuromuscular and functional adaptations (16,61).

It is worth noting that, because RT is commonly incorporated as a complement to the field-specific training, it seems that soccer players would benefit from RT programs based on low-to-moderate levels of intraset fatigue because they would produce higher

performance improvements at a lower time and fatigue (13,37) cost. Nonetheless, it is important to consider that training to failure (or very close to it) has also been found as an effective strategy to generate positive adaptations on strength, but especially on muscle mass gain (16,54). Consequently, this type of stimuli could represent an interesting strategy during specific periods included in soccer plans in which muscle hypertrophy could be the main objective, such as postinjury recovery or preseason phases.

CONTRIBUTIONS OF THE VELOCITY-BASED TRAINING

The fact that the above-mentioned review studies focused only on a dichotomous question (i.e., to reach the muscle failure or not) limited to some extent the practical applications of their findings. For example, considering that a player could perform ~ 19 repetitions against 65% of his 1RM in the full squat exercise (20), the single instruction based on whether reaching or not the muscle failure lacks sufficient concision to accurately program the training stimulus.

Alternatively, different programming and dosing methodologies have emerged from the VBT, which goes beyond whether to reach or not the

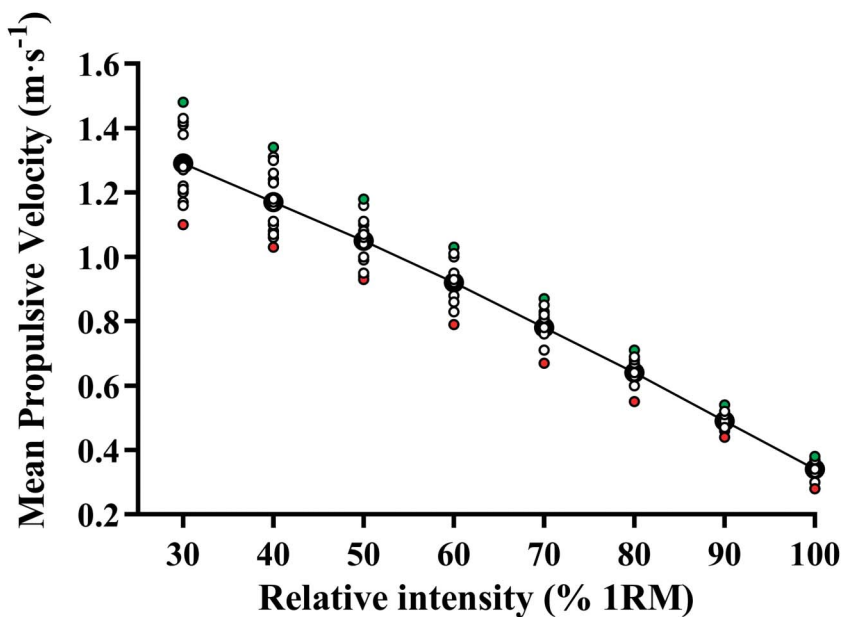


Figure. An example of possible between-subject differences (each point is a player) in mean propulsive velocity attained at relative intensities from 30 to 100% 1RM in a non-paused full squat exercise. Green-filled and red-filled points represent the players that achieved the maximal and minimal velocities at each %1RM, respectively. Black-filled points represent the mean of all velocities at each %1RM.

muscle failure. These approaches are defined and illustrated with examples in Table 3, which also includes their advantages and disadvantages. According to previous studies in soccer, programming RT with lower levels of effort (e.g., ~15% of velocity loss in each set or complete 5–6 repetitions from 12 to 14 possible) has been shown effective to induce improvements in neuromuscular performance (39). Incurring lower fatigue during RT may help in a faster recovery (40,59), which may be critical when dealing with weekly competitions. Thus, both VBT and alternative approaches (Table 3) seem operative methods to avoid overtraining and program individualized resistance exercises in soccer.

VELOCITY-BASED TRAINING AS AN EVALUATION METHOD: PRETRAINING-POSTTRAINING EFFECTS AND FATIGUE STATUS

TRADITIONAL METHODOLOGIES

Similar to traditional programming strategies, conventional RT interventions

including soccer players have used the 1RM test as the main evaluation to measure changes in strength (9,57,60). In this incremental test, the player lifts progressively heavier and heavier absolute loads (without velocity measurements) to the heaviest load he or she can lift without any external help and completing the full ROM of the exercise (i.e., reaching the 1RM) (4). Although the 1RM test has been used for decades as the main assessment of the dynamic strength, it only informs about a single point of the force-velocity relationship (15). Thus, this parameter would lack information about the ability of the player to apply force to the other loads included within the force-velocity spectrum, especially to those who have to deal with on the field (41).

CONTRIBUTIONS OF THE VELOCITY-BASED TRAINING

In addition to incorporating sport-specific evaluations like sprints or jumps, the limitation of the traditional 1RM test to evaluate performance changes against light-moderate loads

can be overcome by measuring pretraining-posttraining changes in barbell velocity. Specifically, considering that the only way to achieve a higher barbell velocity against the same absolute load (in kilograms) is to exert more force (if the execution technique has remained identical) (2), practitioners could add the barbell velocity to their testing routine to evaluate the changes in dynamic strength across the entire spectrum of loads (27,35,39,44,46).

Furthermore, practitioners could estimate the “strength deficit” using the L-V relationship. The strength deficit is defined as the difference between the force applied against the 1RM load (i.e., 100% 1RM) and that applied against any other %1RM (25). In a practical context, modifications in the strength deficit can be evaluated by comparing the changes in MPV against each %1RM after a RT intervention. Thus, the higher MPV the player achieves against each %1RM after the training program, the smaller the strength deficit with that %1RM. In turn, in practice, a lower strength deficit could translate into the player will be able to apply more force at that % 1RM (2).

On the other hand, because barbell velocity is a direct consequence of the force applied, changes in barbell velocity against the same absolute load could be used as an indicator of the fatigue status of the player. Specifically, the change in the load that elicits a $1.00 \text{ m} \cdot \text{s}^{-1}$ has been proposed as a valid and practical strategy to evaluate the neuromuscular recovery status (32,37). First, the validity of this strategy would be supported by the high relationship that has been found between the changes in the load that elicits $1.00 \text{ m} \cdot \text{s}^{-1}$ and metabolic fatigue products such as the lactate (bench press, $r = 0.97$; full squat, $r = 0.93$) and ammonia (bench press, $r^2 = 0.95$; full squat, $r^2 = 0.90$) (47). Furthermore, the reduction of the velocity against the load lifted at $1.00 \text{ m} \cdot \text{s}^{-1}$ in the full squat exercise was found to be closely associated ($r = 0.93$) with

Table 3
Strategies to program and dose the intraset fatigue in soccer players

	Methodology		
	Level of effort (20)	Velocity loss (VL) (14)	Effort index (EI) (45)
Definition	The number of reps performed in the set (volume) concerning the maximum number of reps that can be completed (intensity)	Monitoring the progressive decline of the barbell velocity over the set of reps	Product of multiplying the velocity reached in the first rep of the set by the percentage of VL over the set of reps
Examples	6 (12): To complete 6 reps with an absolute load that would allow the player to perform a total of 12 (~75% 1RM)	15% VL: to reach a barbell velocity of $0.49 \text{ m}\cdot\text{s}^{-1}$ when the first rep of the set was performed at $0.58 \text{ m}\cdot\text{s}^{-1}$	12.2 EI (a.u.): to reach a barbell velocity of $0.61 \text{ m}\cdot\text{s}^{-1}$ in the first rep of the set and lose 20% of velocity over the set
	4 (8): To complete 4 reps with an absolute load that would allow the player to perform a total of 8 (~80% 1RM)	30% VL: to reach a barbell velocity of $0.54 \text{ m}\cdot\text{s}^{-1}$ when the first rep of the set was performed at $0.77 \text{ m}\cdot\text{s}^{-1}$	13.8 EI (a.u.): to reach a barbell velocity of $0.46 \text{ m}\cdot\text{s}^{-1}$ in the first rep of the set and lose 30% of velocity over the set
Advantages	No technology is required (free method) It allows to individualize the training load during group training (e.g., RT programs in team sports) Low interplayer and intraplayer variability It avoids stopping the training set because of lack of concentration of the player (common for the VL and EI methods)	Accurate monitoring of the intraset fatigue It allows programming with high precision the intraset fatigue by prescribing a certain VL or EI, instead of a fixed number of reps with an absolute load The EI permits the comparison of stimuli performed using a different % 1RM or VL. For example: $26.9 \text{ EI: first rep of the set} = 0.77 \text{ m}\cdot\text{s}^{-1} (\sim 60\% \text{ 1RM in BP}) \times 35\% \text{ VL}$ $26.9 \text{ EI: first rep of the set} = 0.49 \text{ m}\cdot\text{s}^{-1} (\sim 80\% \text{ 1RM in BP}) \times 55\% \text{ VL}$	
Disadvantages	It requires a prior familiarization of the player to maximize the accuracy of the method Lower reliability in novice players	Need for highly reproducible technologies and protocols Difficult to be implemented during group training (e.g., RT programs in team sports) Need for a trained coach with extensive knowledge in the proper implementation of the protocols and in the use of the different technologies High-time cost for data processing and analysis It requires a prior familiarization of the player to perform all repetitions at the maximal intended velocity	
Rep = repetition; %1RM = relative intensity; RT = resistance training; VL = velocity loss; EI = effort index; BP = bech press.			

another indicator of mechanical fatigue like the loss of jump capacity (47). Second, it would represent a practical approach because it only involves performing 2–3 repetitions against a moderate load (~45% 1RM in the bench press and ~60% 1RM in the non-paused full squat) (Table 1), which could be performed at the end of the warm-up included in the training session. Based on the information resulting from this test, physical trainers and coaches could make individualized decisions “on the go”, such as giving

rest to a player or reducing his or her workload.

APPLICATION OF VELOCITY-BASED TRAINING: TECHNOLOGIES, VARIABLES, AND TECHNIQUES

The advances provided by the VBT are, however, dependent on the level of reproducibility and repeatability exhibited by the different technologies used for its implementation. In practical terms, the technical error of these velocity-monitoring devices must be inferior to the expected changes to ensure that variations in barbell

velocity are because of actual strength changes and not to technological errors. For instance, an increment in MPV of just $0.07\text{--}0.10 \text{ m} \cdot \text{s}^{-1}$ is associated with improvements of ~5% 1RM strength in the main resistance exercises (Table 1). Therefore, technologies showing a technical error $\geq 0.10 \text{ m} \cdot \text{s}^{-1}$ would not allow practitioners to be confident that the changes measured after a training intervention are the result of an actual increase or decrease in the player's performance (7,30).

Available technologies provide real-time feedback of kinematic metrics

such as the mean velocity (MV: mean concentric velocity from the start of the concentric phase until the bar reaches the maximum height), MPV (described above), peak velocity (PV: maximum instantaneous velocity value reached during the concentric phase), and ROM. Whereas MV and MPV have shown excellent values of reliability (10), some authors have suggested using MPV to increase the sensitivity of the assessment. Briefly, because MV includes the braking phase (the portion of the concentric phase during which the athlete decelerates the barbell), this variable underestimates and overestimates the dynamic strength of stronger and weaker players, respectively (29,48). This fact would occur mainly against light and moderate loads, during which stronger athletes achieve higher velocities, and therefore greater braking phases, than weaker ones (2).

In addition to reproducible and repeatable technologies, velocity-based assessments would benefit from (a) providing visual feedback to the athlete to maximize his or her performance (23) and (b) precise control of the execution technique. Regarding the second aspect, it has been proved that imposing a momentary pause (~2 seconds) between the eccentric and concentric phases (i.e., avoiding the stretch-shortening cycle) produces more reliable velocity results during incremental loading tests (34) and allows players to precisely replicate the individual eccentric ROM in each repetition (28,29). In turn, the ROM has been shown to significantly influence the resulting velocity (the MPV increases as the ROM increases) (28,29). Another aspect that would influence the velocity-based evaluations is the degree of freedom offered in the exercise (i.e., free weight or machine-based training). Specifically, the use of a Smith machine has been shown to increase the reliability values of the strength measurements (22), probably because of the absence of movement in the horizontal plane. Furthermore, the L-V relationship of a specific exercise has been proved to be

influenced by the degree of freedom used to execute it. For example, a reduction of the velocity attained at each %1RM has been found when comparing the L-V relationship of the bench press exercise performed using a weight stack machine (MPV against light loads, <50% = $1.06 \text{ m}\cdot\text{s}^{-1}$; MPV against moderate loads, 50–80% = $0.67 \text{ m}\cdot\text{s}^{-1}$) or a Smith machine (MPV against light loads, <50% = $1.17 \text{ m}\cdot\text{s}^{-1}$; MPV against moderate loads, 50–80% = $0.71 \text{ m}\cdot\text{s}^{-1}$) (43).

In practice, machine-based training allows players to control the execution of each repetition, which could be interesting in certain moments of the session (e.g., warm-up), phases of the season (e.g., rehabilitation after an injury), and for specific objectives (e.g., to learn a complicated technical execution). However, because the agonist-synergist-antagonist complex works in a coordinated manner during most of the movements (i.e., intermuscular coordination) (24), the strength gains generated during machine-based training may find it difficult to be transferred to soccer actions like jumping, accelerations, or changes of direction (63).

PRACTICAL APPLICATIONS AND FUTURE PERSPECTIVES

The strength capacity is a critical determinant of physical performance in soccer. The VBT is a step forward to individually program the target intensity and the level of intraset fatigue to be incurred and to accurately determine the chronic and acute changes in strength. In particular, the VBT approach may assist coaches in (a) setting and adjusting the training load by measuring the barbell velocity and (b) estimating the fatigue incurred to avoid overtraining by considering the loss in velocity during the set. Despite its applications, this method might present limitations for daily-basis use, such as the need for specific equipment, extensive knowledge in the proper implementation of this methodology, and a considerable deal of time to register and

analyze all repetitions in each set. Furthermore, athletes are required to be familiarized and committed to executing each lift at the maximum intended velocity; otherwise, the reliability of the adjustments will decrease. Some practical alternatives, such as the “level of effort” based on the self-assessment of perceived effort, may help in reducing the costs and time required to apply the VBT. Finally, the tool provided in this article (see Supplementary Material, Supplemental Digital Content, <http://links.lww.com/SCJ/A340>) may assist coaches in data collection and interpretation for a daily-basis use.

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